Rozwiązania wybranych przykładów z Z04

$$(x^p)' = px^{p-1}, \quad p \in R$$

$$(\ln x)' = \frac{1}{x}, \qquad (\log_a x)' = \frac{1}{x}\ \frac{1}{\ln a}$$

$$(e^x)' = e^x, \qquad (a^x)' = a^x \cdot \ln a$$

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x, \qquad (\operatorname{tg} x)' = \frac{1}{\cos^2 x}$$

$$(arc\sin x)' = \frac{1}{\sqrt{1-x^2}}, \qquad (arc\cos x)' = \frac{-1}{\sqrt{1-x^2}},$$

$$(arc\operatorname{tg} x)' = \frac{1}{1+x^2}, \qquad (arc\operatorname{ctg} x)' = \frac{-1}{1+x^2}$$

1. $(cf') = cf', \quad (f \pm g)' = f' \pm g'$
2. $(f \cdot g)' = f'g + fg' \quad$ *pochodna iloczynu*
3. $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2} \quad$ *pochodna ilorazu*
4. $\big(f(y(x))\big)' = f'(y) \cdot y'(x) \quad$ *pochodna funkcji złożonej*
5. $f''(x) = \big(f'(x)\big)', \quad f^n(x) = \big(f^{n-1}(x)\big)' \quad n-ta\ pochodna$

0. *a*) $\lim\limits_{x\to\frac{\pi}{7}} \frac{\sin(x)-\sin\left(\frac{\pi}{7}\right)}{x-\frac{\pi}{7}} = a$. Widzimy, ze jest to definicja pochodnej funkcji $\sin x$, w punkcie $x_0 = \frac{\pi}{7}$, dlatego $a = \cos\left(\frac{\pi}{7}\right)$

b) Niech $x_0 = 1$ należy do dziedziny $f(x)$. Jeśli $\lim\limits_{x\to 1} \frac{f(x)-3}{x-1} = 3$, wyznacz $f'(1)$. 1. Zauważamy, że $f(1) = 3$, Z definicji pochodnej otrzymamy, że $f'(1) = 3$.

1. Oblicz pochodne proste (tzn. bez arytmetyki pochodnych) następujących funkcji:

$$(x^p)' = px^{p-1}, \quad p \in R$$

a) $(x^4)' = 4x^3$

b) $(7)' = 0$

c) $(3s)' = 3$

d) $(x^{-7})' = -7x^{-8} = -\frac{7}{x^8}$

$e)\ \ (\sqrt{t})' = \left(t^{\frac{1}{2}}\right)' = \frac{1}{2}t^{-\frac{1}{2}} = \frac{1}{2\sqrt{t}}$

$f)\ \ (\sqrt[4]{x})' = \left(x^{\frac{1}{4}}\right)' = \frac{1}{4}x^{\frac{-3}{4}} = \frac{1}{4}\frac{1}{\sqrt[4]{x^3}}$

$g)\ \ \left(\sqrt[3]{x^4}\right)' = \left(x^{\frac{4}{3}}\right)' = \frac{4}{3}x^{\frac{1}{3}} = \frac{4}{3}\sqrt[3]{x}$

$h)\ \ \left(\frac{1}{y}\right)' = (y^{-1})' = -y^{-2} = \frac{-1}{y^2}$

$i)$ $\boldsymbol{f'(y) = 0}$. funkcja f jest stała względem y. W tym przypadku x jest stałą, również $1/x^3$ jest stałą.

$j)\ \ \left(\frac{1}{\sqrt[3]{x^2}}\right)' = \left(x^{-\frac{2}{3}}\right)' = -\frac{2}{3}x^{-\frac{5}{3}} = -\frac{2}{3}\frac{1}{\sqrt[3]{x^5}}$

$k)\ \left(\frac{a}{a^8}\right)' = \left(\frac{1}{a^7}\right)' = (a^{-7})' = -7a^{-8} = -\frac{7}{a^8}$

$l)\ \ \left(\frac{x^4}{\sqrt[3]{x^8}}\right)' = \left(x^{4-\frac{8}{3}}\right)' = \left(x^{\frac{4}{3}}\right)' = \frac{4}{3}x^{\frac{1}{3}} = \frac{4}{3}\sqrt[3]{x}$

2. Wyznacz pochodne następujących funkcji.

$b)\ f'(t) = \left(\frac{\ln t}{e^t}\right)' = \frac{(\ln t)'e^t - (\ln t)(e^t)'}{(e^t)^2} = \frac{\frac{1}{t}e^t - (\ln t)e^t}{(e^t)^2} = \frac{\frac{1}{t} - \ln t}{(e^t)}$

$c)\quad f'(z) = \left(\frac{1}{\sqrt[3]{2z} + z^2}\right)' = \left(-\frac{1}{\left(\sqrt[3]{2z} + z^2\right)^2}\right)\left(\frac{1}{3}(2z)^{-\frac{2}{3}} \cdot 2 + 2z\right).$

$d)\quad f'(\alpha) = \left(\frac{1}{\sqrt[3]{2\alpha} + x\alpha^2}\right)' = \left(-\frac{1}{\left(\sqrt[3]{2\alpha} + x\alpha^2\right)^2}\right)\left(\frac{1}{3}(2\alpha)^{-\frac{2}{3}} \cdot 2 + 2x\alpha\right).$

$g)\ \ f'(y) = \left(5\tan\frac{y}{5} + \tan\frac{\pi}{8}\right)' = 5\frac{1}{\cos^2\left(\frac{y}{5}\right)} \cdot \frac{1}{5} + 0 = \frac{1}{\cos^2\left(\frac{y}{5}\right)}$

$h)\quad f'(x) = \left(\sqrt{x + \sqrt{x + \sqrt{x}}}\right)' = \frac{1}{2\sqrt{x + \sqrt{x + \sqrt{x}}}} \cdot \left(1 + \frac{\left(1 + \frac{1}{2\sqrt{x}}\right)}{2\sqrt{x + \sqrt{x}}}\right)$

$f'(x) = \left(sin^2\left(\frac{1 - \ln x}{x}\right)\right)' = 2\,sin\left(\frac{1 - \ln x}{x}\right)cos\left(\frac{1 - \ln x}{x}\right)\left(\frac{-\frac{1}{x}x - (1 - \ln x)}{x^2}\right)$

$k)\ \ f'(x) = \left(\frac{1}{\sqrt{x}} e^{(x^2+ln\,x+1)}\right)' = \left(\frac{e^{(x^2+ln\,x+1)}}{\sqrt{x}}\right)' = \frac{e^{(x^2+ln\,x+1)}\left(2x+\frac{1}{x}\right)\sqrt{x} - e^{(x^2+ln\,x+1)}\frac{1}{2\sqrt{x}}}{x} =$

$$= \frac{e^{(x^2+1)}x\left[\left(2x+\frac{1}{x}\right)\sqrt{x} - \frac{1}{2\sqrt{x}}\right]}{x} = e^{(x^2+1)}\left[\left(2x+\frac{1}{x}\right)\sqrt{x} - \frac{1}{2\sqrt{x}}\right]$$

$l)\ \ f'(x) = \left(e^{sin\,x}\right)' = e^{sin\,x}\,cos\,x$

3*. Oblicz pochodne [**wsk**: $x = e^{\ln x}$]

$a)\quad y = x^x;$

$$(x^x)' = \left(e^{\ln(x^x)}\right)' = \left(e^{x\cdot\ln(x)}\right)' = e^{x\cdot\ln(x)}(x\cdot\ln(x))' = e^{x\cdot\ln(x)}\left(\ln x + x\frac{1}{x}\right)$$
$$= e^{x\cdot\ln(x)}(\ln x + 1) = x^x(\ln x + 1)$$

b) $y = \sin^x x$

$$f'(x) = ((sin\,x)^x)' = \left(e^{ln((sin\,x)^x)}\right)' = \left(e^{x\,ln(sin\,x)}\right)' = e^{x\,ln(sin\,x)}\left(ln(sin\,x) + x\cdot\frac{cos\,x}{sin\,x}\right) =$$

$$= e^{x\,ln(sin\,x)}\,(ln(sin\,x) + x\cdot cot\,x) \quad = (sin\,x)^x\ \ (ln(sin\,x) + x\cdot ctgx)$$

$c)\ \ y = x^{sin\,x}$

$$f'(x) = \left(x^{sin\,x}\right)' = \left(e^{ln(x^{sin\,x})}\right)' = \left(e^{sin\,x\,ln\,x}\right)' = e^{sin\,x\,ln\,x}\left(cos\,x\,ln\,x + sin\,x\cdot\frac{1}{x}\right) =$$
$$= x^{\sin x}\left(\cos x\ln x + \sin x\cdot\frac{1}{x}\right)$$

e) $f'(x) = \left((\cos^3 x)^{\cot x}\,e^{2x}\right)' = \left((\cos^3 x)^{\cot x}\right)'e^{2x} + \left(e^{2x}\right)'(\cos^3 x)^{\cot x} =$

$$\left(e^{ln\left((cos^3 x)^{cot\,x}\right)}\right)'e^{2x} + 2e^{2x}(cos^3 x)^{cot\,x} = \left(e^{cot\,x\,ln(cos^3 x)}\right)'e^{2x} + 2e^{2x}(cos^3 x)^{cot\,x} =$$

$$= \left(e^{cot\,x\,ln(cos^3 x)}\right)\left(-\frac{1}{sin^2 x}ln(cos^3 x) + cot\,x\frac{1}{cos^3 x}\cdot 3\,cos^2 x\cdot(-sin\,x)\right)e^{2x}$$
$$+ 2e^{2x}(cos^3 x)^{cot\,x} =$$

$$= (cos^3 x)^{cot\,x}e^{2x}\left(-\frac{ln(cos^3 x)}{sin^2 x} - 3 + 2\right) = -(cos^3 x)^{cot\,x}e^{2x}\left(\frac{ln(cos^3 x)}{sin^2 x} + 1\right)$$

$f)\ f(y) = (\cos y)^{\frac{1}{y}}$

$$f'^{(y)} = \left((\cos y)^{\frac{1}{y}}\right)' = \left(e^{\ln\cos y\cdot\frac{1}{y}}\right)' = e^{\ln\cos y\cdot\frac{1}{y}}\left(\ln\cos y\cdot\frac{1}{y}\right)'$$

$= e^{\ln\cos y \cdot \frac{1}{y}} \left(\frac{-\sin y}{\cos y} \frac{1}{y} + \ln\cos y \frac{-1}{y^2} \right).$

4. Oblicz y'' lub $\frac{d^2y}{dx^2}$ dla

a) $y = x\sqrt{1+x^2}$

$$f'(x) = \left(x\sqrt{1+x^2}\right)' = \sqrt{1+x^2} + x\frac{1}{2\sqrt{1+x^2}} \cdot 2x = \sqrt{1+x^2} + \frac{x^2}{\sqrt{1+x^2}} = \frac{1+2x^2}{\sqrt{1+x^2}}$$

$$f''(x) = \left(\frac{1+2x^2}{\sqrt{1+x^2}}\right)' = \frac{4x\sqrt{1+x^2} - \left(1+2x^2\right)\cdot\frac{1}{2\sqrt{1+x^2}}\cdot 2x}{1+x^2} = \frac{4x\sqrt{1+x^2} - \left(1+2x^2\right)\cdot\frac{x}{\sqrt{1+x^2}}}{1+x^2} =$$

$$= \frac{4x\sqrt{1+x^2} - \left(1+2x^2\right)\cdot\frac{x}{\sqrt{1+x^2}}}{1+x^2} = \frac{4x\left(1+x^2\right) - \left(1+2x^2\right)\cdot x}{\left(1+x^2\right)\sqrt{1+x^2}} = \frac{4x+4x^3-x-2x^3}{\left(1+x^2\right)\sqrt{1+x^2}} = \frac{2x^3+3x}{\left(1+x^2\right)\sqrt{1+x^2}}$$

b) $y = e^{(-x^2)}$

$$f'(x) = \left(e^{(-x^2)}\right)' = e^{(-x^2)} \cdot (-2x) = -2x \cdot e^{(-x^2)}$$

$$f''(x) = \left(-2x \cdot e^{(-x^2)}\right)' = -2e^{(-x^2)} - 2xe^{(-x^2)} \cdot (-2x) = -2e^{(-x^2)} + 4x^2e^{(-x^2)}$$
$$= e^{(-x^2)}(4x^2 - 2)$$

5. Niech $f(x)$ będzie dwukrotnie różniczkowalne. Oblicz y'' czyli $\frac{d^2y}{dx^2}$

a) $y' = \left(f(x^2)\right)' = f'(x^2)2x,$ $\qquad y'' = (f'(x^2)2x)' = f''(x^2) \cdot 2x \cdot 2x + 2f'(x^2).$

$$y''' = \left(2 \cdot f'(x^2) + 4x^2 \cdot f''(x^2)\right)' = 2 \cdot f''(x^2) \cdot 2x + 8x \cdot f''(x^2) + 4x^2 \cdot f'''(x^2) \cdot 2x =$$

$$= 12x \cdot f''(x^2) + 8x^3 \cdot f'''(x^2)$$

b) $y = f(1/x)$

$$y' = \left(f(1/x)\right)' = -\frac{1}{x^2} \cdot f'(1/x)$$

$$y'' = \left(f(x^{-1})\right)'' = (-f'(x^{-1}) \cdot x^{-2})' = f''(x^{-1}) \cdot x^{-2} \cdot x^{-2} + f'(x^{-1})2x^{-3} \quad \text{lub}$$
$$y'' = \left(-\frac{1}{x^2} \cdot f'(1/x)\right)' = \frac{2}{x^3} \cdot f'(1/x) + \left(-\frac{1}{x^2}\right) \cdot \left(-\frac{1}{x^2}\right) \cdot f''(1/x) = \frac{2}{x^3} \cdot f'(1/x) + \frac{1}{x^4} \cdot f''(1/x)$$

$$y''' = \left(\frac{2}{x^3} \cdot f'(1/x) + \frac{1}{x^4} \cdot f''(1/x)\right)' =$$

$$-\frac{6}{x^4}f'(1/x)+\frac{2}{x^3}\cdot f''(1/x)\cdot\left(-\frac{1}{x^2}\right)+\left(-\frac{4}{x^5}\right)\cdot f''(1/x)+\frac{1}{x^4}\cdot f'''(1/x)\cdot\left(-\frac{1}{x^2}\right)=$$

$$-\frac{6}{x^4}f'(1/x)-\frac{6}{x^5}\cdot f''(1/x)-\frac{1}{x^6}\cdot f'''(1/x)$$

c) $y=f(\ln x)$

$$\big(f(\ln x)\big)''=\left(f'(\ln x)\frac{1}{x}\right)'=f''(\ln x)\frac{1}{x^2}-f'(\ln x)\frac{1}{x^2}.$$

Równanie prostej stycznej do $f(x)$ w punkcie $(x_0,\ f(x_0))$:

$$\boldsymbol{y=f(x_0)+f'(x_0)(x-x_0)}$$

6. Wyznacz równanie stycznej do wykresu podanej funkcji f (tzn. liniowe przybliżenie tej funkcji f) we wskazanym punkcie.

b) $f(x)=2x^2+3x-7$ at $(-1,-8)$ $D_f = R$

$f'(x)=(2x^2+3x-7)'=4x+3$

$f'(-1)=-4+3=-1$

$y=-8-1(x+1)$

$y=-8-x-1$ zatem $y=-x-9$

Odp.: Równanie stycznej do wykresu $f(x)=2x^2+3x-7$ w $(-1,-8)$ to jest $y=-x-9$.

7. a) Napisz równanie stycznej do krzywej $y=x^2+2x$ w punkcie $(1;3)$.
$x_0=1$

$f(x_0)=3$

$f(x)=x^2+2x$

$f'(x)=2x+2$

$f'(x_0)=4$

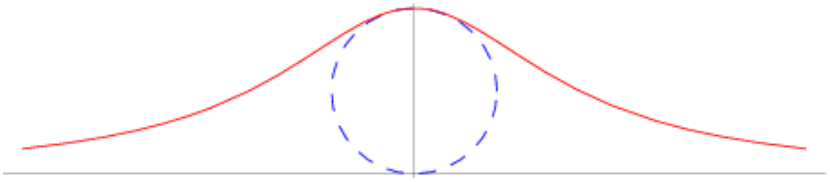
$y=3+4(x-1)$

$y=3+4x-4$

$y=4x-1$

Odp..: Styczna do wykresu $f(x)=x^2+2x$ w $(1,3)$ to $y=4x-1$

b) $f(x)=\frac{1}{1+x^2}$, $x_0=-1$, $f(x_0)=\frac{1}{2}$



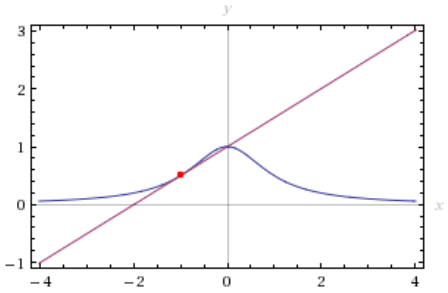
$$f'(x)=-\frac{2x}{(1+x^2)^2}$$

$$f'(-1)=\frac{1}{2}$$

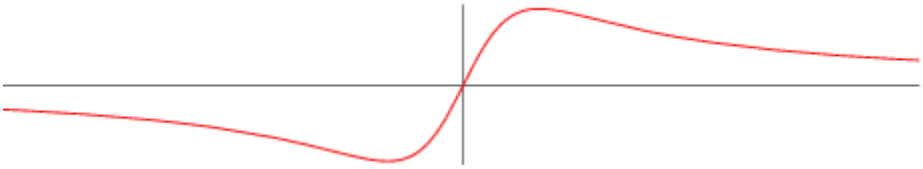
$$y=\frac{1}{2}+\frac{1}{2}(x+1) \quad \Rightarrow \quad y=\frac{1}{2}+\frac{1}{2}x+\frac{1}{2} \quad \Rightarrow \quad y=\frac{1}{2}x+1$$

Odp..: Styczna do wykresu $f(x)=\frac{1}{1+x^2}$ w $(-1,1/2)$ to $y=\frac{1}{2}x+1$.

WolframApha: tangent to 1/(1+x^2) at x=-1



c) *Serpentyna* $f(x)=\frac{x}{x^2+1}, x_0=-3, \qquad f(x_0)=-0.3$



$$f'(x)=\frac{x^2+1-x\cdot 2x}{(x^2+1)^2}=\frac{1-x^2}{(x^2+1)^2}$$

$$f'(-3)=-\frac{8}{100}=-\frac{2}{25}$$

$y=-0.3-\frac{2}{25}(x+3)=-\frac{2}{25}x-\frac{27}{50}$ - styczna.